SEMICON India Hackathon 2026 — Idea Submission

EPDR-Net

Edge-Preserving Degradation-Adaptive Deep Restoration Network

for Semiconductor Inspection Images

SEMICON India Hackathon 2026 — Problem Statement: AI-Based Restoration of Degraded Images for Semiconductor Inspection

1. Team Details

Sr. No.	Role	Name	Academic Year
1	Team Leader	M.KAVYA	2026-27
2	Member 1	T.ANITHA BLESSY	2026-27
3	Member 2	S.PUJITHA	2026-27
4	Member 3	[Enter Name]	[Enter Year]

Team Name: NANOVISION

College Name: ADITYA UNIVERSITY

Team Leader Contact: 8106653563

Team Leader Email: makireddikavya2008@gmail.com

2. Problem Statement Addressed

AI-Based Restoration of Degraded Images for Semiconductor Inspection. SEM inspection images are frequently degraded by speckle noise, Gaussian noise and reduced spatial resolution — often simultaneously. This causes loss of critical structures such as metal lines, gate edges, vias, contacts and FinFET fins, leading to inspection uncertainty and missed defect detection. A restoration model must generalize to out-of-distribution (OOD) noise, geometry and resolution while running fast enough for real inspection pipelines.

3. Idea Description

EPDR-Net is a lightweight, edge-aware deep learning model that restores noisy, low-resolution SEM images into clean, high-resolution ones for reliable inspection. Rather than treating this as generic image denoising, EPDR-Net is purpose-engineered around semiconductor inspection: it has two complementary branches — a Content Restoration branch that removes speckle/Gaussian noise and reconstructs missing spatial detail, and an Edge Preservation branch that explicitly learns semiconductor feature boundaries (metal lines, gate edges, vias, contacts, trenches, FinFET fins, DRAM word/bit lines). These are combined using a Degradation-Adaptive Fusion Module, guided by an estimate of the noise, blur and resolution level present in the input.

4. Proposed Solution

The end-to-end pipeline: the degraded grayscale SEM image is normalized, then passed through a Degradation Estimation Module that outputs a degradation vector $z = [z_{speckle}, z_{gaussian}, z_{resolution}]$. This vector conditions a Degradation-Adaptive Fusion Module that combines features from the Content Restoration branch (a lightweight residual encoder built from depthwise-separable convolutions) and the Edge Preservation branch (a Sobel/Laplacian-driven edge encoder). Fused features pass through Multi-Scale Feature Reconstruction (operating at 1x, 1/2x and 1/4x scales) and PixelShuffle upsampling (x2 or x4). Residual learning is used throughout: the network predicts a residual added to a bicubic-upsampled base image, rather than reconstructing the image from scratch, which makes training more stable.

Training uses a composite loss combining Charbonnier loss (robust reconstruction), an edge loss on Sobel maps (structure preservation), SSIM loss (perceptual structural similarity), and an FFT-domain frequency loss (preserving periodic DRAM/FinFET patterns): $L = 0.40 \cdot L_{char} + 0.25 \cdot L_{edge} + 0.20 \cdot L_{SSIM} + 0.10 \cdot L_{freq} + 0.05 \cdot L_{perc}$. A two-stage strategy trains first on individual degradations (Gaussian, speckle, low-resolution) and then on combined, harder degradations to build OOD robustness.

5. System Architecture

The figure below shows the complete EPDR-Net pipeline: degradation estimation, the parallel Content Restoration and Edge Preservation branches, degradation-adaptive attention fusion, multi-scale reconstruction, and residual-corrected output.

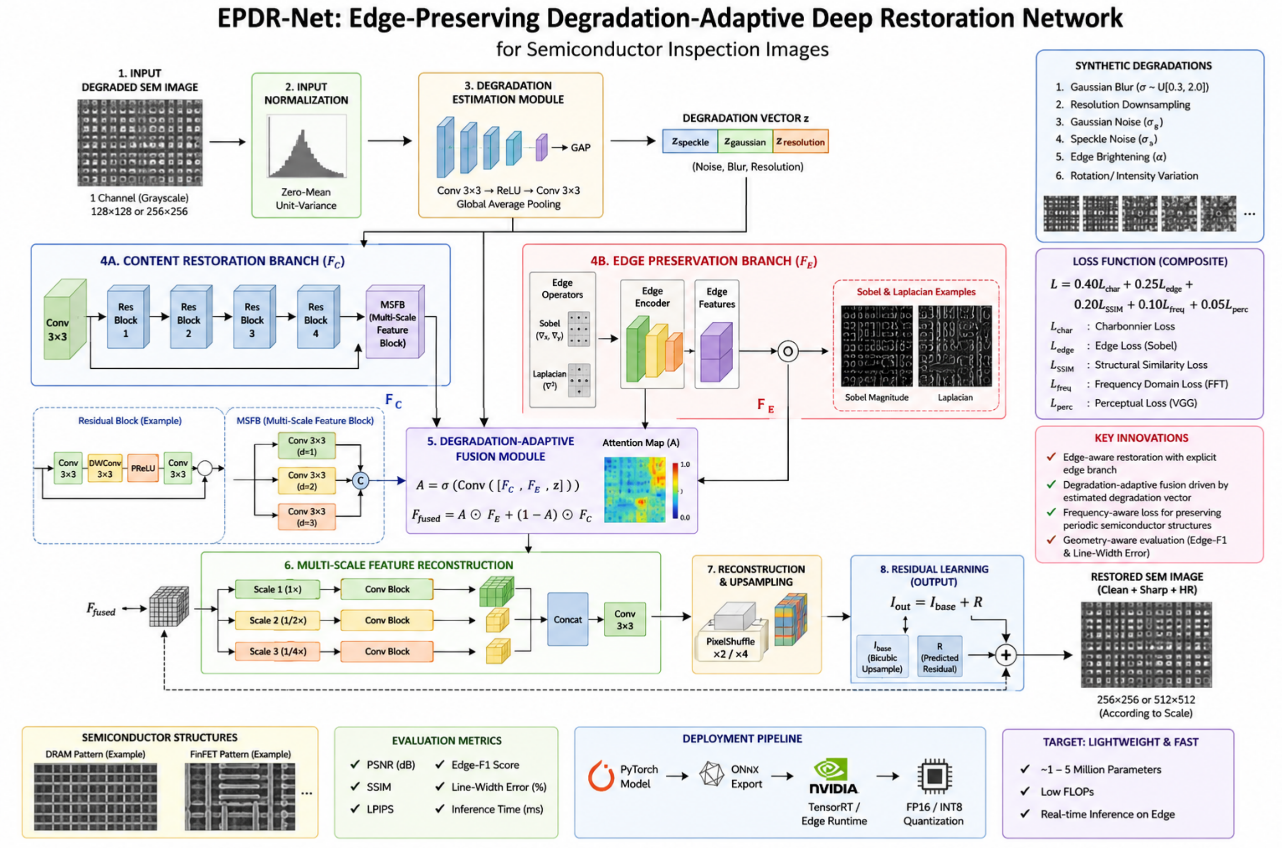

Figure 1: EPDR-Net end-to-end architecture for SEM image restoration.

6. Innovation and Uniqueness

- **Edge-aware restoration** — a dedicated Sobel/Laplacian-driven branch stops denoising from destroying line, gate and via boundaries.
- **Degradation-adaptive fusion** — the network estimates the noise/blur/resolution level and dynamically reweights how it restores each image.
- **Frequency-aware loss (FFT)** — protects the periodic DRAM/FinFET structures that plain pixel losses tend to blur out.
- **Geometry-aware evaluation** — beyond PSNR/SSIM/LPIPS, Edge-F1 and Line-Width Error score how well critical geometry is preserved, which is what inspection actually depends on.

Unlike generic denoisers (DnCNN, SRCNN, U-Net) built for natural images, EPDR-Net is engineered for SEM inspection: it stays lightweight (1–5M parameters, depthwise-separable convolutions) for real-time edge inference, generalizes to OOD noise/geometry/resolution, and is judged on whether it preserves the exact geometry that inspection and yield decisions depend on — not just visual quality.

7. Impact and Benefits

Primary Impact: Restores clean, high-resolution SEM images from noisy/low-resolution captures in real time, reducing missed or misclassified defects and giving inspection teams reliable geometric detail (line widths, edges, contacts) for yield and defect decisions on the fab floor.

Quantifiable Outcomes: Target model size of 1–5M parameters (versus 100M+ transformer baselines). Reports PSNR, SSIM, LPIPS, Edge-F1 and Line-Width Error across in-distribution and OOD (noise/geometry/resolution) test sets, plus FP32/FP16/INT8 latency benchmarks for edge deployment.

8. Technology & Feasibility / Methodology Used

Software: Python, PyTorch, OpenCV, NumPy, scikit-image, CUDA. Architecture: dual-branch CNN (Content Restoration + Edge Preservation) with a Degradation Estimation module and attention-based adaptive fusion, trained with a composite Charbonnier + Edge + SSIM + Frequency + Perceptual loss on a synthetic DRAM/FinFET degradation dataset. Deployment: PyTorch ONNX → TensorRT, quantized to FP16/INT8 for real-time inference on edge inspection hardware.

Feasibility: kept to 1–5M parameters and depthwise-separable convolutions specifically for low-latency, low-memory edge deployment.

Component Details

Software: Python, PyTorch, OpenCV, NumPy, scikit-image, CUDA

Architecture: Dual-branch CNN + Degradation Estimation + Adaptive Fusion

Hardware Components: GPU (training); Edge inference hardware (NVIDIA Jetson-class / TensorRT runtime)

Development Tools: PyTorch, OpenCV, CUDA, ONNX, TensorRT

9. Dataset & Training Strategy

Synthetic dataset generation is central to this project: DRAM-style patterns (word lines, bit lines, contacts/vias) and FinFET-style patterns (fins, gates, gate intersections) are generated programmatically, then degraded with a physically motivated pipeline — Gaussian blur (σ [0.3, 2.0]), resolution downsampling (e.g., 512→256 or 256→128), additive Gaussian noise, multiplicative speckle noise, and optional edge brightening for SEM-like appearance. Data is split at the structure level (e.g., DRAM-A/B for training, DRAM-C for validation) rather than randomly per image, to test genuine structural generalization.

Training proceeds in two phases: single-degradation pretraining, followed by combined hard-degradation fine-tuning (high noise + strong blur + strong downsampling + rotation + intensity variation) to build OOD robustness.

11. Research and References

Research Background & Methodology: Built on established restoration principles — residual encoder-decoder CNNs for image restoration, Sobel/Laplacian edge operators for structure-preserving loss design, Charbonnier loss for robustness to outliers, SSIM for structural fidelity, and FFT-domain loss for periodic-pattern preservation — combined into a degradation-adaptive, edge-aware framework purpose-built for semiconductor SEM inspection imagery.

- Ref 1: Dong et al., “Image Super-Resolution Using Deep Convolutional Networks” (SRCNN), IEEE TPAMI 2016
- Ref 2: Zhang et al., “Beyond a Gaussian Denoiser: Residual Learning of Deep CNN for Image Denoising” (DnCNN), IEEE TIP 2017
- Ref 3: Ronneberger et al., “U-Net: Convolutional Networks for Biomedical Image Segmentation,” MICCAI 2015